



Semester III

Molecular Biology

Project

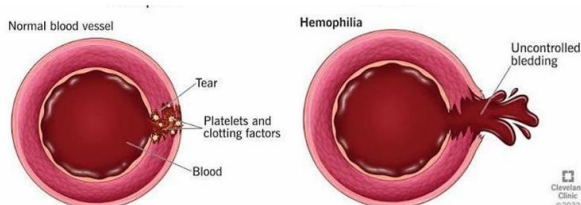
HAEMOPHILIA

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What is Haemophilia?

Haemophilia is a disorder of blood clotting. Normally, when somebody injures themselves, the blood clots in a few minutes and wound healing can begin. In haemophilia, this doesn't happen. This is because one of the ingredients needed for blood clotting doesn't work properly. This deficiency in the activity of the ingredient may be complete or partial. When the deficiency is complete, the person is said to have severe haemophilia. Even in severe haemophilia, clotting is not prevented altogether. Instead, it is much slower than normal. The result is a friable clot that cannot withstand the pressure of escaping blood. Body tissues involved become disrupted, and wounds do not heal properly. In severe haemophilia, most bleeds are internal, into joints and muscles. With modern treatment given early, there is often nothing to see and no interference with everyday life.



Source : <https://my.clevelandclinic.org/health/diseases/14083-hemophilia>

Normal vessels form clots using platelets and clotting factors, stopping bleeding. In hemophilia, missing clotting factors prevent clot formation, causing uncontrolled bleeding.

What causes Haemophilia?

When a person bleeds, the body typically pools the blood cells together to form a clot to stop bleeding. Clotting factors are proteins in the blood that work with cells known as platelets to form clots. Haemophilia occurs when a clotting factor is missing or the level of clotting factor is low.

Congenital haemophilia

Haemophilia is usually inherited, meaning a person is born with the disorder (congenital). Congenital haemophilia is classified by the type of clotting factor that's low. The most common type is haemophilia A, associated with a low level of factor 8. The next most common type is haemophilia B, associated with a low level of factor 9.

Acquired haemophilia

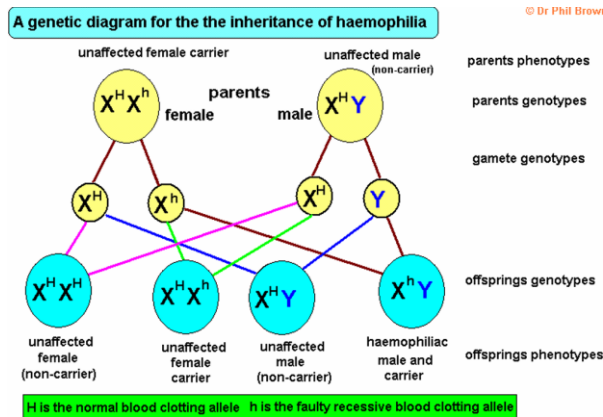
Some people develop Haemophilia with no family history of the disorder. This is called acquired Haemophilia. Acquired Haemophilia is a variety of the condition that occurs when a person's immune system attacks clotting factor 8 or 9 in the blood. It can be associated with:

- Pregnancy
- Autoimmune conditions
- Cancer
- Multiple sclerosis
- Drug reactions

Haemophilia inheritance

In the most common types of Haemophilia, the faulty gene is located on the X chromosome. Everyone has two sex chromosomes, one from each parent. Females inherit an X chromosome from the mother and an X chromosome from the father. Males inherit an X chromosome from the mother and a Y chromosome from the father.

This means that Haemophilia almost always occurs in boys and is passed from mother to son through one of the mother's genes. Most women with the defective gene are carriers who have no signs or symptoms of Haemophilia. But some carriers can have bleeding symptoms if their clotting factors are moderately decreased.



Source: <https://www.docbrown.info/ebiology/genome6-3.htm>

X^H = normal blood-clotting allele; X^h = recessive haemophilia allele.
Diagram shows inheritance when a carrier mother ($X^H X^h$) mates with an unaffected father ($X^H Y$)

Risk Factor

The biggest risk factor for Haemophilia is to have family members who also have the disorder. Males are much more likely to have Haemophilia than are females.

Types of Haemophilia

There are several types of haemophilia: haemophilia A, haemophilia B, haemophilia C, parahaemophilia, acquired haemophilia A, and acquired haemophilia B.

Haemophilia A

Haemophilia A is a blood clotting disorder caused by a genetic deficiency in clotting factor VIII, thereby resulting in significant susceptibility to bleeding, both internally and externally. This condition occurs almost exclusively in males born to carrier mothers due to X-linked recessive inheritance. Nevertheless, rare isolated cases do emerge from de novo (spontaneous) mutations. The medical management of individuals with haemophilia A frequently entails the administration of factor VIII medication through slow intravenous injection. This intervention aims to address and pre-empt additional bleeding episodes in affected individuals.

Haemophilia B

Haemophilia B, also spelled haemophilia B, is a blood clotting disorder causing easy bruising and bleeding due to an inherited mutation of the gene for factor IX, and resulting in a deficiency of factor IX. It is less common than factor VIII deficiency (haemophilia A). Haemophilia B was first recognized as a distinct disease entity in 1952. It is also known by the eponym Christmas disease, named after Stephen Christmas, the first patient described with haemophilia B. Most individuals who have haemophilia B and experience symptoms are men. The prevalence of haemophilia B in the population is about one in 40,000; haemophilia B represents about 15% of patients with haemophilia. Many female carriers of the disease have no symptoms. However, an estimated 10-25% of female carriers have mild symptoms; in rare cases, female carriers may have moderate or severe symptoms. Symptoms include easy bruising, urinary tract bleeding (haematuria), nose bleeds (epistaxis), and bleeding into joints (haemarthrosis).

Haemophilia C

Haemophilia C (also known as plasma thromboplastin antecedent (PTA) deficiency or Rosenthal syndrome) is a mild form of haemophilia affecting both sexes, due to factor XI deficiency. It is the fourth most common coagulation disorder after von Willebrand's disease and haemophilia A and B. In the United States, it is thought to affect 1 in 100,000 of the adult population, making it 10% as common as haemophilia A.

Parahaemophilia

The type of haemophilia known as parahaemophilia is a mild and rare form and is due to a deficiency in factor V. This type can be inherited or acquired.

Acquired Haemophilia A

A non-genetic form of haemophilia is caused by autoantibodies against factor VIII and so is

known as acquired haemophilia A. It is a rare but potentially life-threatening bleeding disorder caused by the development of autoantibodies (inhibitors) directed against plasma coagulation factors. Acquired haemophilia can be associated with cancers, autoimmune disorders and following childbirth.

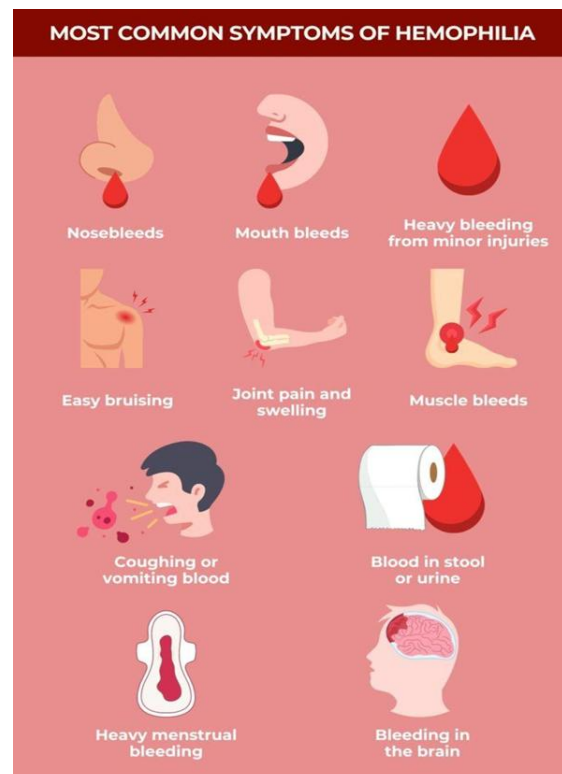
MAIN TYPES OF HEMOPHILIA			
	Definition	Missing clotting factor	Prevalence
Hemophilia A	 Genetic bleeding disorder caused by mutations in the <i>F8</i> gene	Factor VIII	• More common in men • 1 in every 5,000 male births
Hemophilia B	 Genetic bleeding disorder caused by mutations in the <i>F9</i> gene	Factor IX	• More common in men • 1 in every 25,000 male births
Hemophilia C	 Genetic bleeding disorder caused by mutations in the <i>F11</i> gene	Factor XI	• Affects men and women equally • 1 in every 100,000 people
Acquired hemophilia	 Autoimmune condition caused by the immune system's misdirected attack against clotting factors	Factor VIII in most cases	• Affects men and women equally; more common in older individuals • 1.5 in every million people

The chart summarizes the four main types of hemophilia, showing their genetic cause, the missing clotting factor, and how common each type is. Hemophilia A and B are X-linked genetic disorders, while Hemophilia C and acquired hemophilia affect both men and women. (source : PWonlyIAS)

Symptoms

Characteristic symptoms vary with severity. In general, symptoms are internal or external bleeding episodes, which are called "bleeds". People with more severe haemophilia experience more severe and more frequent bleeds, while people with mild haemophilia usually experience more minor symptoms except after surgery or serious trauma. In cases of moderate haemophilia symptoms are variable which manifest along a spectrum between severe and mild forms. In both haemophilia A and B, there is spontaneous bleeding but a normal bleeding time, normal prothrombin time, normal thrombin time, and prolonged partial thromboplastin time. Internal bleeding is common in people with severe haemophilia and some individuals with

moderate haemophilia. The most characteristic type of internal bleed is a joint bleed where blood enters into the joint spaces. This is most common with severe haemophiliacs and can occur spontaneously (without evident trauma). If not treated promptly, joint bleeds can lead to permanent joint damage and disfigurement. Bleeding into soft tissues such as muscles and subcutaneous tissues is less severe but can lead to damage and requires treatment. Children with mild to moderate haemophilia may not have any signs or symptoms at birth, especially if they do not undergo circumcision. Their first symptoms are often frequent and large bruises and haematomas from frequent bumps and falls as they learn to walk. Swelling and bruising from bleeding in the joints, soft tissue, and muscles may also occur. Children with mild haemophilia may not have noticeable symptoms for many years. Often, the first sign in very mild haemophiliacs is heavy bleeding from a dental procedure, an accident, or surgery. Females who are carriers usually have enough clotting factors from their one normal gene to prevent serious bleeding problems, though some may present as mild haemophiliacs.



The image illustrates the most common symptoms of hemophilia, including excessive bleeding, bruising, joint pain, and internal bleeding. (source: Hemophilianewstoday.com)

Diagnosis

The diagnosis of haemophilia is a critical first step in managing this lifelong genetic bleeding disorder. Since the condition is caused by a deficiency in specific clotting factors—Factor VIII (Haemophilia A) or Factor IX (Haemophilia B)—accurate identification is essential for initiating factor replacement therapy and preventing potentially life-threatening hemorrhagic episodes. The diagnostic journey typically involves a multi-step approach, beginning with a detailed clinical history and progressing through a series of specialised laboratory coagulation tests.

Brief History on the Diagnosis of Haemophilia

2nd Century AD : The Talmud, a collection of Jewish rabbinical writings on laws and traditions, stated that if two brothers died from circumcision, the next son should not undergo the procedure, suggesting an early understanding of a hereditary bleeding disorder.

1803-1900 : John Conrad Otto, a Philadelphia physician, was the first to publish an article recognizing that a hemorrhagic bleeding disorder primarily affected men, and ran in certain families further in 1820 Friedrich Nasse was a German physician expanded on this, correctly stating that carrier mothers could pass the trait to their sons.

Mid 20th Century : Identification of different types of Haemophilia Type A caused by factor VIII and Type B by factor IX.

Late 20th Century : The development of blood-clotting tests, such as the activated partial thromboplastin time (APTT), allowed for the measurement of specific clotting factors like VIII and IX, making diagnosis more precise.

Present : Today, diagnosis relies on both a patient's family and medical history, along with laboratory tests that measure specific clotting factor levels and molecular genetic testing is also used to identify specific genetic variants in the F8

Types of Tests

Screening Tests (General Coagulation)

TEST NAME	WHAT IT MEASURES	TYPICAL RESULT IN HEMOPHILIA
Activated Partial Thromboplastin Time (aPTT)	How long it takes for a clot to form using a specific pathway (involving factors VIII, IX, XI, and XII).	Prolonged (longer than normal) in Hemophilia A and B.
Prothrombin Time (PT)	How long it takes for a clot to form using a different pathway (involving factors I, II, V, VII, and X).	Normal in most cases of Hemophilia A and B

Table 'a'. The table compares clotting tests (aPTT and PT) by showing what each measures and how results typically appear in hemophilia. aPTT is prolonged in Hemophilia A and B, while PT usually remains normal

Clotting Factor Tests (Factor Assays)

Factor VIII in the blood. A low level confirms Haemophilia A.

Factor IX Assay: Measures the level and activity of Factor IX in the blood. A low level confirms Haemophilia B.

The result from these assays determines the severity:

- **Severe:** Less than 1% of normal factor activity.
- **Moderate:** 1% to 5% of normal factor activity.
- **Mild:** Greater than 5% but less than 40% of normal factor activity

Additional Diagnostic & Monitoring Tests:

Mixing Study: Done if the aPTT is prolonged. It involves mixing the patient's blood with normal plasma. If the clotting time corrects, it confirms a factor deficiency. If it doesn't correct, it

suggests the presence of an inhibitor (an antibody that blocks the clotting factor). Bethesda Assay: Specifically measures the level of inhibitors (antibodies) against Factor VIII or IX, which can develop in some patients as a complication of treatment.

Genetic Testing (DNA Testing)

This test examines the F8 gene (for Haemophilia A) or the F9 gene (for Haemophilia B) to identify the specific genetic mutation responsible for the condition

When?

In High-Risk Newborns (Soon After Birth) :

- If there is a known family history of haemophilia (e.g., the mother is a known carrier), testing is often done immediately or shortly after birth.
- Umbilical Cord Blood Testing: A blood sample from the umbilical cord can be used for factor level or genetic testing.
- Symptom-Driven: Testing is urgent if a newborn shows signs like:
 - Prolonged bleeding after circumcision.
 - Excessive bleeding after heel sticks (for newborn screening tests).
 - Intracranial (head/brain) bleeding after a difficult deliver

During Infancy and Early Childhood (Symptom Onset):

- If there is no known family history, the diagnosis is usually made when bleeding symptoms become noticeable.
- Severe Haemophilia: Most cases of severe Haemophilia (very low clotting factor levels) are diagnosed in the first year of life (median age is often cited as around 1 month), as infants begin to crawl and move, leading to:

- Unexplained bruising and large hematomas (lumps of blood).
- Spontaneous bleeding into joints (hemarthroses) or muscles.
- Moderate Haemophilia: Usually diagnosed in early childhood due to bleeding following minor injuries, or occasional spontaneous bleeding

Later in Childhood or Adulthood (Mild Haemophilia)

- People with mild Haemophilia (clotting factor levels are higher) may not be diagnosed until later in life, as they typically only bleed after significant challenges to the clotting system.
- Women/Girls: Testing is also warranted for women/girls (who are usually carriers but can have symptoms) who experience:
 - Heavy or prolonged menstrual bleeding (menorrhagia).
 - Excessive bleeding after childbirth or surgery.
- Surgery or Trauma: Diagnosis may happen after prolonged or excessive bleeding following:
 - Major surgery
 - Serious injury or accident
 - Tooth extraction or other dental procedures.

Treatment

Haemophilia treatment is critical, moving beyond acute care to ensure long term health and a near-normal quality of life. Its immediate objective is to halt life-threatening hemorrhages (like intracranial bleeds) and swiftly resolve existing acute muscle and joint bleeds using factor concentrates. Crucially, modern care emphasizes prophylaxis—regular factor administration—which dramatically

prevents recurrent bleeding into the joints. This proactive approach is essential for averting chronic joint damage, such as hemophilic arthropathy, which causes long-term pain and disability. Effective treatment also manages complex complications, notably the development of Inhibitors. Ultimately, consistent therapy transforms Haemophilia from a debilitating, high-risk condition into a manageable chronic disorder, significantly improving life expectancy and activity levels.

Brief History of Treatment of Haemophilia.

Mid-20th Century: Treatment primarily consisted of whole blood or fresh plasma transfusions, which contained very small amounts of the missing clotting factors and often required hospitalization. Life expectancy remained low.

Clotting Factor Era (1960s-1980s):

Cryoprecipitate (Mid-1960s): A major breakthrough came in 1964 with the discovery of cryoprecipitate by Dr. Judith Graham Pool. This was a frozen fraction of plasma rich in Factor VIII (the factor missing in Haemophilia A). It was a more concentrated treatment that significantly improved a patient's ability to manage bleeds

Factor Concentrates (1970s): Freeze-dried (lyophilized) plasma derived Factor VIII and Factor IX concentrates became widely available. This revolutionized care because it allowed for home treatment (patients could self-infuse), enabling faster response to bleeds and the initiation of prophylaxis (regular infusions to prevent bleeds). This drastically improved life expectancy and quality of life.

Late 20th Century:

Recombinant Factors (Late 1980s–1990s): The ultimate evolution in replacement therapy was the introduction of recombinant factor concentrates (not derived from human blood). Approved for use starting in the late 1980s and

early 1990s, these products became the standard of care, offering a genetically engineered and highly pure clotting factor.

Extended Half-Life (EHL) Factors (2010s): Further engineering created EHL factors, which are modified to last longer in the bloodstream. This innovation allows patients to maintain adequate clotting levels with fewer and less frequent intravenous infusions for preventative care.

Present History

Non-Factor Therapies: The newest class of treatments includes non factor replacement therapies, such as the bispecific antibody emicizumab. This is administered via an easy, subcutaneous injection and works by mimicking the function of the missing clotting factor (Factor VIII), simplifying the prophylaxis regimen.

Gene Therapy: Current progress is focused on gene therapy, which aims to offer a single-infusion, potentially curative treatment for both Haemophilia A and B by delivering a correct copy of the factor gene to the patient's liver cells.

Types of Treatment

Factor Replacement Therapy (The Standard of Care): This involves replacing the missing clotting factor (Factor VIII for Haemophilia A, Factor IX for Haemophilia B) by infusing it directly into a vein.

Recombinant Factors: These are genetically engineered, lab-made factors that are not derived from human blood. They are the most widely used and standard type of factor concentrate.

Extended Half-Life (EHL) Factors: These are newer, modified recombinant factors designed to last longer in the body.

- **Benefit:** They allow for less frequent infusions (e.g., once or twice a week instead of multiple times a week) for prophylaxis

(preventative treatment), which significantly improves quality of life and adherence.

- Newer Ultra-Long Half-Life (UHL) factors are also emerging, which can maintain near-normal factor levels for even longer periods.

Administration: These can be given as prophylaxis (regularly to prevent bleeds) or on demand (to stop an active bleeding episode)

Non-Factor Therapies (Novel Agents): These treatments correct the bleeding defect without directly replacing the missing factor protein. They aim to rebalance the coagulation system

- Bispecific Antibody (Factor VIII Mimetic):
 - The most well-known example is emicizumab (for Haemophilia A).
 - It's a bispecific antibody that mimics the function of Factor VIII, bridging two other clotting factors to help the blood clot.
 - Benefit: It is administered via a simple subcutaneous injection (under the skin) rather than an intravenous infusion, and often less frequently (weekly, bi-weekly, or monthly).
- Hemostatic Rebalancing Agents:
 - This class of drugs works by targeting and inhibiting natural anti-clotting proteins (like antithrombin or Tissue Factor Pathway Inhibitor) to boost the body's overall clotting potential.
 - Examples include treatments like Fitusiran or Concizumab, which are typically given by subcutaneous injection.

Gene Therapy (The Potential Cure) Gene therapy aims to be a one-time treatment that

provides a potentially long lasting or even curative solution.

- Mechanism: A working copy of the Factor VIII or Factor IX gene is delivered into the body (usually targeting the liver cells) using a vector (often a modified virus). The liver cells then start producing the missing clotting factor.
- Goal: To raise the patient's baseline clotting factor levels high enough so that they no longer require regular factor infusions or have a significantly reduced number of bleeds.
- Current Status: Several gene therapies for both Haemophilia A and B have received regulatory approval and are available for eligible adult patients.

Other Supporting Treatments:

- Desmopressin (DDAVP): A synthetic hormone used for mild Haemophilia A that stimulates the release of Factor VIII stored in the body's blood vessel linings.
- Antifibrinolytic Agents (e.g., Tranexamic acid): Medications that help preserve and stabilize a formed blood clot, often used for mouth or nose bleeds, or before dental procedures.
- Bypassing Agents (for Inhibitors): Used to stop bleeding in patients who have developed inhibitors (antibodies that destroy the infused factor) and include products like activated Prothrombin Complex Concentrates (aPCCs).

Future Prospects of Therapy

Gene Therapy Viral vectors such as the adeno-associated viruses (AAV) can alleviate the deficiencies associated with hemophilia and so they provide a permanent solution. These viruses enable the body to create its own clotting factors and this reduces the number of infusions a patient might require. With these gene therapies for example, Etranacogene

Dezaparovec has been effective in restoring the concentration of factor IX in B hemophiliacs. Regional Accessibility It is important to note there is a variation in the global access of advanced therapies. For instance, Turkey is one of the countries that is trying_ to adopt gene therapy in the usual treatment while dealing with issues such as finances and the availability of clinics.^{3,11}. [\(19\)](#)

Tragedy of Clotting factor era.

The Haemophilia Tainted Blood Tragedy of the mid-1980s was a global disaster where thousands of people with Haemophilia were infected with HIV (leading to AIDS) and Hepatitis C (HCV). The primary cause was the use of Factor Concentrate, a life-saving blood clotting product. This treatment was derived from pooled plasma from up to 40,000 donors, often including high-risk, paid individuals. If even one donation was contaminated, the entire batch was ruined. Despite warnings in the early 1980s that AIDS was blood-borne, the medical and regulatory communities were slow to implement critical safety measures like heat treatment to kill viruses or donor screening. This delay resulted in catastrophic infection rates. In the US, for example, nearly half of those with severe Haemophilia were infected with HIV, and over 90% were infected with HCV in some countries, leading to the deaths of thousands and a subsequent legacy of suffering, legal battles, and public inquiries worldwide.

Little Statistics About a Large Tragedy:

Metric	Approximate Statistics (Varies by Source/Country)
Hemophilia Patients Infected with HIV	In the US, nearly 50% of people with severe hemophilia became infected with HIV (some sources cite up to 90% in certain subgroups).
Hemophilia Patients Infected with HCV	In many countries, including the US and UK, a very high percentage, often over 90% , of those treated with concentrate between 1970-1985 were infected with Hepatitis C .
Deaths (Overall Tainted Blood)	Worldwide, the overall death toll from HIV and Hepatitis C infections is in the thousands . For the UK, over 3,000 people have died as a result of the scandal.
Children Infected	Hundreds of children with hemophilia were infected with HIV, often leading to shortened lives .

The table summarizes global statistics on HIV and HCV infections, deaths, and affected children linked to contaminated blood products in hemophilia patients.

Data vary by country but highlight the severe historical impact of tainted clotting-factor treatments.

Living with Haemophilia

Nine members of The Haemophilia Society took part in a semi-structured interview. The interviews were tape recorded and transcribed and the results subjected to thematic content analysis.

Physical Impact

All interview participants reported that the bleeding into joints associated with their condition had a major impact on physical well-being. Bleeding episodes in the joints were reported to be the cause of 'trigger joints' which are susceptible to joint damage and are also the joints in which participants had developed arthritis. Participants with arthritis emphasised how the condition limited their mobility and independence. As a result of these bleeds into joints, four participants had had joint replacements. One participant was expecting to have a replacement shortly and had concerns about his role as a father, particularly his ability to hold and cuddle his young baby because of the impact of bleeds on his elbow joint.

In addition to dealing with the impact of their bleeding disorder, six participants reported that they were living with, or had previously lived with a co-morbid condition (e.g. HIV,

Hepatitis C, and arthritis). Participants reported that the treatments for HIV and Hepatitis C had very unpleasant side effects. For example, one participant reported experiencing gastric problems, fatigue and headaches as side effects of medication for HIV. A second participant, also taking medication for HIV, reported exhaustion and weight loss as side effects of medication.

Impact on Social activities, Education and Work.

Participants reported that physical limitations and restrictions had impacted on many areas of their lives including social activities, school and work. For one participant, blood loss had a major impact on social activities and often resulted in time off work.

Three participants reported negative effects on their education. These were associated with earlier, less advanced treatment regimens focused on stopping bleeds once they occurred rather than prevention. Thus, these participants had spent a great deal of time in hospital. For five participants, their condition had impacted on their working lives. As with many other long-term health conditions such as arthritis, multiple sclerosis and diabetes, three participants reported taking early retirement due to their bleeding disorder or other co-morbid conditions arising as a result of their bleeding disorder. For one participant, the perception was that the diagnosis of a virus resulting from treatment for a bleeding disorder, subsequently lead to a loss of promotion and loss of employment.

Three participants commented on the financial impact of giving up work on their lives and that of their families. Loss of employment inevitably leads to a loss of earnings and therefore families have to manage on severely reduced finances. Two participants reported that loss of employment and earnings lead to feelings of dependency as they were no longer the main 'bread winner' but were reliant on a family member. One participant experienced a

change of role to that of a househusband as a consequence of his health condition.

The majority of participants reported that their condition impacted not only on themselves, but also on other members of the family. Most of the participants were diagnosed during childhood and were aware of the impact on their parents of having a child with a bleeding disorder. Participants highlighted that their condition(s) impacted on their parents' working lives and financial circumstances both in the past, while participants were still children, and also at the present time during retirement. Participants felt responsible for the nature of the parents' lives.

One participant reported that due to the genetic nature of haemophilia, their children were also affected and that for some people this could have implications on decisions about whether to have children.

Discrimination

Several participants reported experiencing a considerable degree of discrimination because of their condition, particularly at work or school. The main problem at school was bullying by other children who did not fully understand their condition. Problems at school were exacerbated by teachers' behaviour as they often did not understand the implications for children and life with a bleeding disorder, and participants reported that teachers were not very sympathetic to their needs.

With regard to work, participants also experienced a lack of understanding by others, both management and colleagues. Participants reported that colleagues did not understand what it meant to have a bleed and the serious impact that this had on their health. Furthermore, they reported that colleagues did not seem to understand that although the injections stopped the bleed, this was not the end of the bleed and that they would need to rest. In addition to this lack of understanding experienced at work, one participant reported feeling ignored by colleagues and attributed

this to his bleeding disorder. He reported that others often avoided close physical contact.

As a result of experiencing discrimination, the majority (6) of participants reported that they did not readily disclose the fact that they had a bleeding disorder or give any information to other people about their condition. Indeed, three participants reported that they made a sustained effort to keep their condition a secret.

In contrast, another participant highlighted the risks associated with failure to disclose this kind of information, both to people with the condition and to others. With regard to individuals with the condition, this participant emphasised safety aspects such as ensuring that factor treatment is available and ready at all times or making sure that they are taken to a hospital that can deal effectively with people with bleeding disorders in an emergency. When considering others, the risk of transmission of viruses was highlighted as an important factor to bear in mind.

Stigma and misconceptions

All participants reported that they encountered some degree of stigmatisation about their condition(s) and that they believed that a lot of stereotypes still exist about people with bleeding disorders. Participants reported feeling that the general population still held a number of misconceptions about people with bleeding disorders. The most common misconception was that individuals with haemophilia, in particular, must have HIV. Participants linked this misconception to the media and advertising campaigns of the 1980s, and the erroneous assumption that 'all men who had haemophilia must be gay'.

Participants highlighted that they were often faced with public misconceptions around the fear of touching and sharing things with an individual with a bleeding disorder.

Participants believed that there is a misunderstanding among the general public

that they might hurt the person with the bleeding disorder by touching them. Furthermore, participants reported that the public think that people with bleeding disorders need protecting and should not be exposed to situations in which they may be harmed. Two participants reported observations made by others such as 'people with haemophilia would eventually become wheelchair users'.

Psychological impact

All participants reported that the symptoms they experienced as a result of their condition(s) and coping with these symptoms on a daily basis had a negative effect on their psychological well-being, particularly anxious and depressed moods. Anxiety about the future was mentioned by five participants as they faced a great deal of uncertainty due to the unpredictable nature of their condition(s). Furthermore, they highlighted the fact that they had to deal with the life-threatening nature of the condition. Concerns for their long-term well-being and the impact of this on the family were also mentioned. Furthermore, participants also reported feeling angry and frustrated because of their condition. Feelings of frustration were often due to being restricted in their daily activities and feelings of anger were often related to viruses such as Hepatitis C, which individuals contracted through infected batches of blood factor which they were treated with. Feelings of depression were related to an individual's inability to do certain activities independently and leading to a sense of 'disempowerment'. Furthermore, symptoms of their condition often restricted individuals who were forced to stay at home. Restricted social contact and feelings of dependence on others exacerbated depressed moods.

Communicating with health professionals

The final theme to emerge was that of difficulties communicating with and being

understood by healthcare practitioners. One participant commented that due to previous negative experiences, he avoided going to his GP and preferred to attend the local Haemophilia Centre.

Problems with Accident & Emergency Departments (A&E) were raised by two participants. One participant reported being turned away by an A&E Department and having to go to another hospital for treatment. A second participant reported delays getting treatment for a joint bleed at A&E, which could result in problems such as arthritis in the long term. One participant did, however, have a positive experience of the NHS Dental Service.⁽¹⁾

Bleeding after vaccination

Vaccination against communicable diseases is crucial for disease prevention, but this practice poses challenges to healthcare professionals in patients with haemophilia. Poor knowledge of the vaccination requirements for these patients and safety concerns often results in vaccination delay or avoidance.

People with haemophilia recommends regular testing for blood-borne infections, such as HIV and hepatitis. They also recommend getting vaccinations for hepatitis A and B. People with haemophilia who receive donated blood products may be at risk of contracting these conditions.

People with haemophilia are at risk of developing hepatitis B from the injection of contaminated concentrates of clotting factor. It is generally recommended that they should be immunised against the virus. Injections of hepatitis B vaccine are usually given intramuscularly into the upper arm. Intramuscular injections, however, are generally considered to be contraindicated in haemophiliacs, some authorities stating that they should not be given in any circumstances. As a result hepatitis B vaccine has been given

to haemophiliacs by subcutaneous³ or intravenous⁴ injection.

Patients, methods, and results

From the survey conducted by **Department of Medicine, Bristol Royal Infirmary**.⁽²⁾

They gave hepatitis B vaccine (H-B-Vax, Merck Sharp and Dohme) to 51 children attending our haemophilia centre (table). The dose, as recommended by the manufacturers, was 0.5 ml (10 µg) for children aged less than 10 years and 1.0 ml (20 µg) for the others. Each child received a course of three injections. Thirty-eight were immunised in the upper arm, 10 in the buttock, and three in the thigh. The injection was given with a 23-gauge needle. Whenever possible pressure was applied to the vaccination site for one to two minutes after the injection. Six of the 153 injections (4%) resulted in bruising, none severe enough to warrant an injection of concentrate. Three children had repeated attacks of vomiting, which began a few hours after the first dose in two and two days after the third dose in the third. One child had a febrile illness with myalgia two days after the third dose of vaccine. There were no other reactions. They did not routinely measure titres of hepatitis B antibody.⁽²⁾

Numbers of children with haemophilia or von Willebrand's disease given hepatitis B vaccine by severity of disease and by age

Condition	Age ≤10 years	Age >10 years	Total
Haemophilia A:			
Severe*	15	9	24
Moderate or mild†	7	3	10
Haemophilia B:			
Severe*	2	1	3
Moderate or mild†	3‡	1	4
Von Willebrand's disease:			
Girls	3	3	6
Boys	2	2	4

*Clotting factor concentration ≤10 U/l.

†Clotting factor concentration >10 U/l.

‡Includes one girl with symptoms.

The table shows how many children with hemophilia A, hemophilia B, or von Willebrand's disease received the hepatitis B vaccine, grouped by age and disease severity. Severity is based on clotting-factor levels, with footnotes explaining factor-concentration cutoffs and special cases. Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC1663310/pdf/bmj00185-0034b.pdf>

Comment

The manufacturer's data sheet for the hepatitis B vaccine has always stated that the vaccine is for intramuscular use only and should not be given intravenously, subcutaneously, or intradermally. We therefore decided to give it by intramuscular injection. Others have given the vaccine subcutaneously³ or intravenously⁴ because of the theoretical risk of bleeding. No differences were reported between haemophiliacs given subcutaneous or intravenous injections and normal controls. Of the 11 haemophiliacs who responded to a questionnaire in a study by Janco,² two (18%) complained of soreness or swelling at the site of injection. In a series of 1083 homosexual men injected intramuscularly the incidence of sore arms was 15.8%. The incidence of bruising in our series was lower (4%). The difference may be due to the fact that in the other series the incidence of side effects was estimated by means of a questionnaire, whereas in our series it was established by routine questions at follow-up visits. Furthermore, we studied only children, who would perhaps be unlikely to complain to their parents about sore arms for fear of a further injection. None was needed—an important fact at a time of anxiety about the safety of clotting factor concentrates. The point we wish to make, however, is that there were no serious bruises and no treatment was needed for haematoma induced by intramuscular injection. The low incidence of side effects is probably due to several factors, including the type of material injected, its volume, and the size of the needle. Other products may not prove to be as free from the risk of causing bruising. Our experience shows that intramuscular injections are no longer completely contraindicated in haemophiliacs.⁽²⁾

Statistics All over the World

Survey 1

From the survey conducted by World Federation of Haemophilia⁽³⁾.

Over a 15-year period (2006–2021), we identified 34 patients with AHA within a Manitoba provincial population ranging from 1,183,981 (2006) to 1,386,333 (2021). The mean population-based incidence rate of AHA was 1.78 ± 0.67 cases per million per year (SD 1.4, range 0–6 cases per year) (Table S1). The number did not vary substantially over the period of data collection. The median age at presentation was 76 years (IQR; 67, 84) with most patients (71%; $n = 24$) presenting in their 7th or 8th decade of life. Females represented 58% (20/34) of cases. Most cases were considered idiopathic (79%; $n = 27$); 18% ($n = 6$) had an underlying autoimmune disease, of which rheumatoid arthritis was most common, and 3% ($n = 1$) had active malignancy at the time of diagnosis. No patients presented in the postpartum period. Most patients (74%; $n = 23$) presented to an emergency department for diagnosis and treatment and 76% ($n = 26$) required admission to hospital. At diagnosis, median FVIII activity was 0 % (IQR; 0, 2) and median inhibitor titre was 30.5 BU/mL (IQR; 15.5, 170).

Data collected on prescribed medications at the time of presentation revealed that two patients were receiving anticoagulants (apixaban $n = 1$, warfarin $n = 1$), and seven patients were receiving antiplatelets (acetylsalicylic acid $n = 6$, clopidogrel $n = 1$). These medications could have impacted bleeding severity and were held at presentation. Signs or symptoms of bleeding were present in 97% ($n = 33$) patients at the time of diagnosis. The remaining patient ($n = 1$) had presented to hospital for an unexplained major bleeding event 4 months prior, but an aPTT was not performed at that time. Major

bleeding events occurred at the time of presentation and diagnosis in 58% ($n = 19$) of patients, and clinically relevant bleeding was present in 15% ($n = 5$). Multiple bleeding sites were identified in 59% ($n = 20$) of patients; the mean number of sites was 2.5 (SD 0.7). Subcutaneous bleeding events were observed in 88% ($n = 29$) patients. Bleeding at critical sites occurred in 24% ($n = 8$) patients, including gastrointestinal ($n = 4$), intracranial ($n = 2$), retroperitoneal ($n = 1$) and one neck hematoma with airway compromise. Other non-critical bleeding sites included nasal, joint and genitourinary.

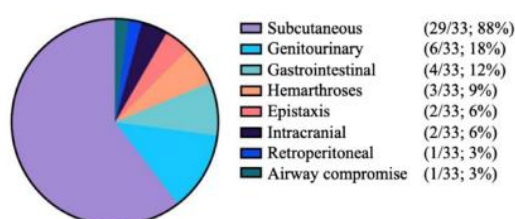


FIGURE 1 Distribution of bleeding sites at presentation from among 33 patients presenting with bleeding at the time of presentation that resulted in a diagnosis of acquired haemophilia A. Multiple sites (mean 2.45 bleeding sites (standard deviation 0.7)) were present in 20 patients.

Source: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/hae.14845>

The diagnosis was delayed in 44% ($n = 15$) of patients, as they reported onset of bleeding symptoms at a median of 4 months (3, 5) prior to diagnosis. Among these patients, the most common bleeding symptom was a history of easy bruising ($n = 13$). Bleeding in critical sites occurred in two patients: one retroperitoneal hematoma and one gastrointestinal bleed with a decrease in haemoglobin of 50 and 58 g/L, respectively. Other non-critical bleeding sites included genitourinary, nasal, and oral mucosa. Laboratory values corresponding to reported bleeding symptoms were found in 11 of the 15 cases, of which nine (82%) did not have an aPTT performed despite abnormal bleeding. The other two patients had an isolated prolonged aPTT of 54 and 59 s, respectively; however, consultation with a haematologist was not initiated.

Haemostatic treatment was administered in 65% ($n = 22$) patients. The median duration of haemostatic treatment was 7 days (IQR; 2, 23). Recombinant FVIIa was the most common first line haemostatic agent, used in 17 patients, whereas activated prothrombin complex concentrate was used in four, and recombinant porcine FVIII in four. Tranexamic acid was used as adjunctive haemostatic treatment in nine patients. Desmopressin was used in three patients, in combination with other haemostatic agents in two patients, and as monotherapy in one patient (Figure 2). The distribution of haemostatic agents use according to FVIII activity and inhibitor titre is shown in Figures S4-5. No thrombotic events were observed during the haemostatic treatment period.

Records capturing IST use were available for 33 patients. IST was administered to 91% ($n = 30$) of patients. Of the three patients who did not receive IST, two were palliated because of terminal comorbid illnesses, and one died of bleeding prior to laboratory confirmation of inhibitors. The combination of steroids (oral prednisone) and oral cyclophosphamide was the most common first line IST regimen, used in 93% ($n = 28$) of patients who received IST. Initial therapy with prednisone alone was used in two patients, although cyclophosphamide was subsequently added in one patient. The remaining patient died during prednisone treatment with a new diagnosis of oesophageal cancer. Of the patients who received combined prednisone and cyclophosphamide, 39% ($n = 11$) required additional lines of treatment. Rituximab was used as second-line therapy in 25% ($n = 7$) of patients, with a median time from IST start to first rituximab dose of 50 days (IQR; 23, 63). Other IST used in second and third-line treatment included azathioprine ($n = 2$) and vincristine ($n = 1$). Median duration of IST course was 136 days.

The most common adverse effects from treatment were cytopenias, occurring in 53% ($n = 16$) patients treated with IST. Lymphopenia was the most common ($n = 12$), followed by

neutropenia ($n = 6$), anaemia ($n = 5$) and thrombocytopenia ($n = 3$). Cytopenias were typically mild to moderate, with the exception of lymphopenia, which was often severe. No infectious complications due to IST were documented during initial hospitalization or in the outpatient bleeding disorders clinic records, where patients were closely followed while on IST. Remission, defined as FVIII activity $> 50\%$ without active bleeding after discontinuation of all haemostatic agents for > 24 h, was achieved in 79% ($n = 27/34$) of patients, of whom 96% ($n = 26/27$) completed IST. IST records were not available for one patient who achieved remission. Median time to remission was 2.1 months (IQR; 1.2, 3.2) (Figure 3). Comparison of time to remission between subgroups, determined by FVIII activity or inhibitor titre at diagnosis, was limited due to the available sample size. Of the patients who achieved remission, two relapsed at 14 and 45 months. Remission was not achieved in seven patients of whom three did not receive IST; two died during IST, and two did not complete IST due to noncompliance or a change in clinical status (Table 2).

TABLE 2 Comparison of outcomes according to remission status.

	Remission	No remission
Total patients (n)	27	7
Received IST (n)	26 ^a	4
Completed IST (n)	26 ^a	0
Time to remission (months)		
Median (IQR)	2.1 (1.2, 3.2)	NA
Relapse (n)	2	NA
Mean time from remission to relapse (months)	29.4 (SD 21.7)	NA
Died as of January 2022 (n)	11/27 (41%)	7/7 (100%)
Time from diagnosis to death (months)		
Median (IQR)	87.2 (14.2, 113.4)	2.6 (0.03, 5.4)

IQR, interquartile range; IST, Immunosuppressive therapy; NA, not applicable; SD, Standard deviation.

^aOne patient achieved remission but no records are available on IST.

status ($n = 2$), malignancy and COVID-19 pneumonia (patient was not receiving IST).

At two years following diagnosis, 10 patients had died, of whom three had achieved remission. Median time from diagnosis to death was 129.5 days (IQR; 19, 332). Median time from diagnosis to death was 332 days

(IQR; not applicable) among the patients who achieved remission, compared to 78 days (IQR; 11, 162) among those who did not. Cause of death was known for two of the three patients who had achieved remission, and included acute leukaemia and cardiac arrest. In the group that did not achieve remission, cause of death was known for five of the seven patients, and included bleeding ($n = 2$), palliative.

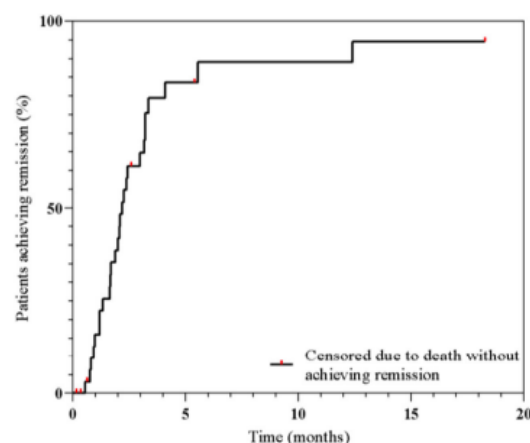


FIGURE 3 Time to remission in all patients. Patients who died without achieving remission are censored. Median time from diagnosis to remission was 2.1 months (IQR; 1.2, 3.2).

Source: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/hae.14845>

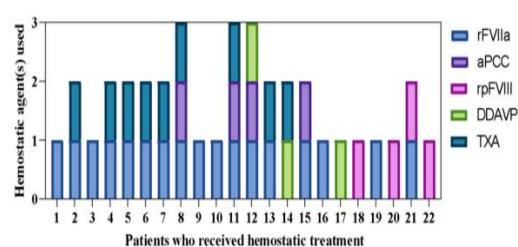


FIGURE 2 Haemostatic agents used in each patient. Patients are listed in chronological order according to the date of diagnosis (April 2006 to February 2021) to illustrate temporal trends in management. Only patients who received haemostatic agents are represented. aPCC, activated prothrombin complex concentrate; DDAVP, desmopressin; rFVIIa, recombinant activated factor VII; rpFVIII, recombinant porcine factor VIII; TXA, tranexamic acid.

Source: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/hae.14845>

refer⁽³⁾

Survey 2

From the data collected by the World Federation of Haemophilia (WFA).⁽⁴⁾

Distribution by Region

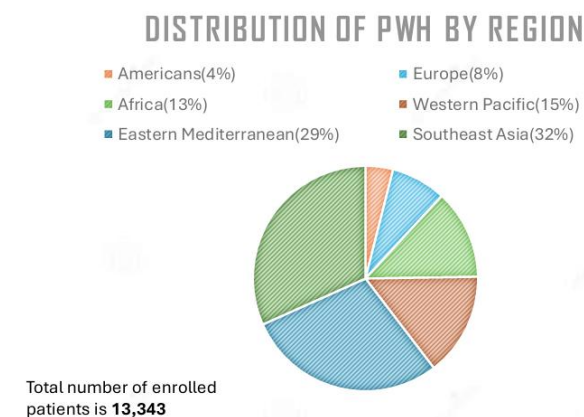


Figure 'a': The chart shows the regional distribution of people with hemophilia (PWH) enrolled in the study, with Southeast Asia and the Eastern Mediterranean contributing the largest shares. A total of 13,343 patients were included across all regions.

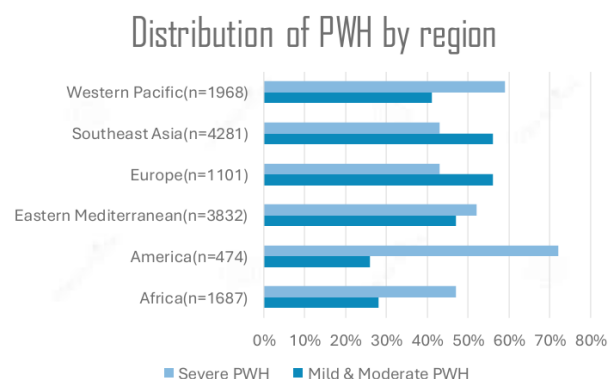
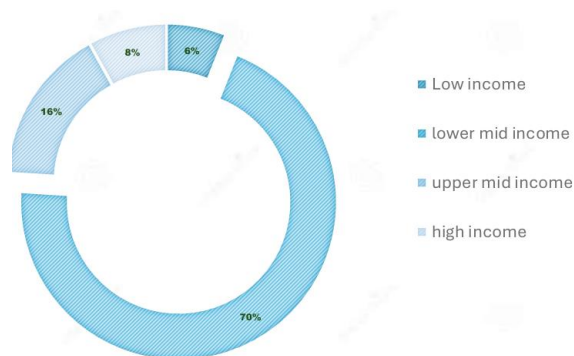


Figure 'b': The light blue bars represent Severe PWH (Persons with Hemophilia). The dark blue bars represent Mild & Moderate PWH (Persons with Hemophilia).

Distribution of PWH by Gross National Income



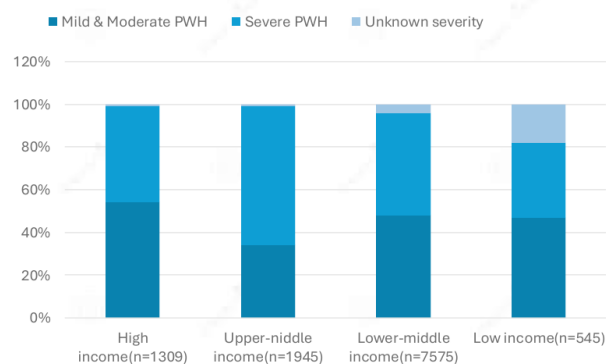
Low income: 6% (Darkest blue)

Lower mid income: 70% (Main blue section)

Upper mid income: 16% (Light blue/gray section)

High income: 8% (Lightest blue/gray section)

Distribution of PWH by GNI



Dark Blue: Mild & Moderate PWH

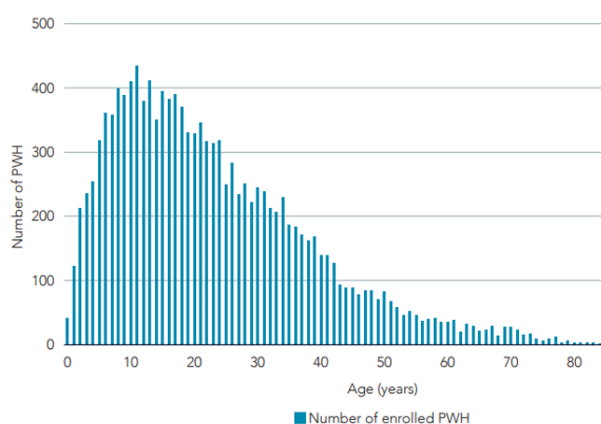
Medium Blue: Severe PWH

Light Blue/White: Unknown severity

Age Distribution of PWH in WBDR

The median age of PWH was 20 years, ranging from 1 month to 98 years.⁽⁴⁾

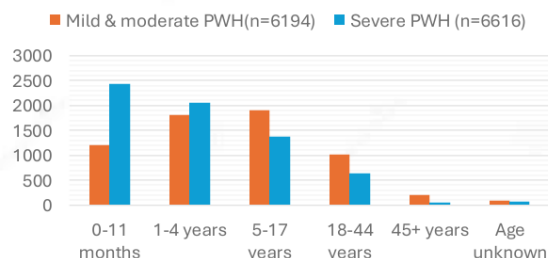
Figure 6
Age distribution of PWH in the WBDR



Source: <https://www1.wfh.org/publications/files/pdf-2452.pdf>

Dark Blue Bars: Number of enrolled PWH (Persons with Hemophilia)

Age at diagnosis by age category



Orange/Red bars: Mild & moderate PWH (n=6194)

Blue bars: Severe PWH (n=6616)

Age at diagnosis by age category

	Mild & Moderate PWH (n=6,194)	Severe PWH (n=6,616)
Age at diagnosis, months, median (IQR)	62(16-172)	20(7-88)
Age at first bleed*, months, median(IQR)	18(6-60)	8(5-18)
Age at first joint bleed**, months, median(IQR)	36(12-74)	18(10-47)

PWH (People With Hemophilia) groups are compared by **median (Interquartile Range, IQR)** age in months for diagnosis, first bleed (*), and first joint bleed (**).

n indicates the sample size for Mild & Moderate PWH (\$n=6,194\$) and Severe PWH (\$n=6,616\$).

*532 PWH with unknown severity were excluded. 2023 Data for 757 PWH from Czechia were not available at the time of publication. ** Includes PWH who reported at least one target joint in 2023; Target joints are defined as '3 or more spontaneous bleeds into a single joint within a consecutive 6 month period. Where there have been ≤ 2 bleeds into the joint within a consecutive 12-month period the joint is no longer considered a target joint

Employment

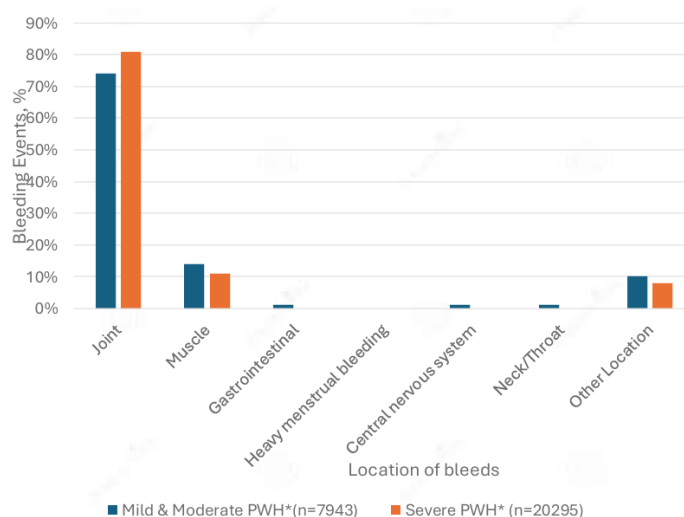
	Mild & Moderate PWH (n=2463)	Severe PWH (n=6,616)
Employed full time or part time	980(40%)	952(36%)
Employed part-time due to haemophilia	187(8%)	232(9%)
Long term sick leave due to haemophilia	23(1%)	24(1%)
Not employed due to haemophilia	131(5%)	231(9%)
Retired due to haemophilia	24(1%)	19(1%)
Student	804(33%)	818(31%)
Other	314(13%)	358(14%)

PWH (People With Hemophilia) groups are compared by their employment and student status, with values shown as the **count** and **percentage (%)** of subjects within each group.

\$n\$ indicates the sample size for Mild & Moderate PWH (\$n=2,463\$) and Severe PWH (\$n=6,616\$).

20% of adults with severe haemophilia report their employment status is negatively affected by haemophilia.

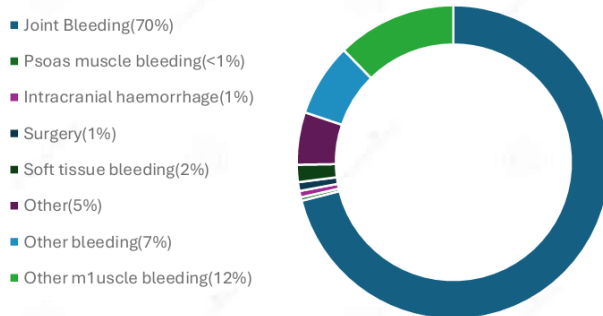
Haemophilia Clinical Data



The table compares the employment and student status of **Mild & Moderate PWH** (n=2,463) and **Severe PWH** (n=6,616) groups.

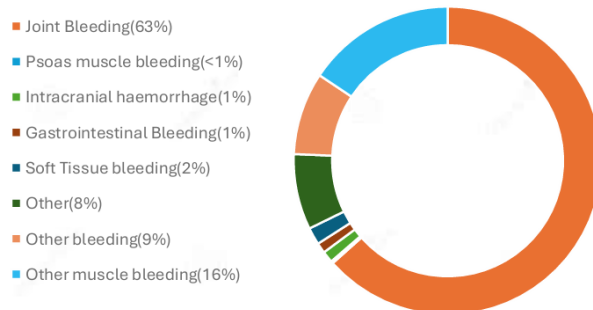
Hospitalisation

Reasons for hospitalisation in Haemophilia A patients



The chart illustrates the frequency of different types of bleeding events in a specific condition (likely Hemophilia), with Joint Bleeding being the overwhelmingly most common event at 70%.

Reasons for hospitalisation in Haemophilia B patients



Joint Bleeding is by far the most common type at 63%, followed by Other muscle bleeding (16%) and Other bleeding (9%). Remaining categories like Intracranial haemorrhage, Gastrointestinal Bleeding, Soft Tissue bleeding, and Psoas muscle bleeding account for only a small percentage.

Summary

In this project, we explored Haemophilia, showing how a single defect in the blood clotting pathway can influence nearly every aspect of a person's life. Whether through inherited mutations in F8 or F9 genes or through acquired autoimmune mechanisms, the project highlighted different types of Haemophilia and how their severity changes clinical symptoms.

We have discussed how crucial early detection is, using tests like aPTT, factor assays, mixed studies, and genetic testing. Historical advances in diagnostic methods have made

identification more accurate, reducing delays that previously led to serious complications.

The project also focuses on treatment evolution: from plasma transfusions to cryoprecipitate, then recombinant factors, EHL products, non-factor therapies like emicizumab, and finally gene therapy, which has shown a potential long-lasting solution.

The project has also addressed past tragedies, especially the contaminated blood crisis, which remains a defining reminder of the importance of safe medical practices and strict regulation. Equally important was the exploration experiences of people with haemophilia — the physical limitations, joint damage, pain, social and educational challenges, stigma, discrimination, and psychological burden.

And data from surveys and registries further demonstrated how haemophilia varies across populations in terms of incidence, age of diagnosis, treatment access, hospitalisation patterns, and socioeconomic impact.

the future of haemophilia treatment is increasingly oriented toward sustained, individualised, and globally accessible care. With advancements in gene therapy, non-factor modalities, and immunological research, the coming decades may witness the transition of haemophilia from a chronic lifelong condition to one with durable, potentially curative solutions and equitable global access, with efforts directed at improving diagnosis, training healthcare personnel, and increasing availability of modern treatments in low- and middle-income countries.

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